

Nandini Gupta

Roll No. 242011019

Technical seminar guide: Dr Swati Chopade

AI-Based Resume Screening for Smart Recruitment: A Fairness-Aware and Explainable LLM-Agent Extension

Abstract

Modern recruitment pipelines must process thousands of resumes per job role, rendering manual screening highly inefficient and prone to subjective human error. While traditional Applicant Tracking Systems (ATS) automate this process, they rely heavily on rigid keyword matching and shallow machine learning models. These legacy systems often fail to capture nuanced semantic context and frequently introduce unintended demographic bias. Recent advancements in Large Language Models (LLMs) have enabled more intelligent, context-aware resume analysis through sophisticated agent-based architectures.

The reference research paper, “*Application of LLM Agents in Recruitment*” by Gan et al., proposes a highly efficient multi-agent architecture in which LLM agents collaboratively perform resume extraction, summarization, and scoring. While this enables intelligent candidate assessment, the resulting scores function as a "black box," lacking the transparency and fairness verification required for ethical HR deployment.

This seminar builds upon that foundational work by proposing a fairness-aware and explainable architectural extension to the existing LLM-agent recruitment framework. The proposed blueprint integrates a counterfactual fairness evaluation module designed to programmatically detect bias arising from sensitive attributes, such as candidate names. Furthermore, it introduces an explainability layer utilizing SHAP values to generate human-readable justifications for assigned candidate scores. By combining TF-IDF features, SBERT embeddings, and an interpretable Logistic Regression scoring engine, the proposed architecture preserves semantic understanding while ensuring complete transparency.

Rather than relying on opaque LLM scoring, this theoretical extension establishes a rigorous, ethically aligned evaluation framework. It enhances trust, accountability, and reliability in AI-driven hiring systems, providing a robust, fair-

minded blueprint suitable for future real-world deployment and empirical testing.
(~250 words)

1. Introduction

Recruitment pipelines today must process large volumes of resumes efficiently while maintaining fairness and consistency. Manual screening is notoriously slow, error-prone, and vulnerable to subjective human bias. While traditional automated Applicant Tracking Systems (ATS) partially address this challenge, they rely primarily on rigid keyword matching, severely limiting their ability to capture deeper semantic meaning.

Recent advancements in transformer-based Large Language Models (LLMs) have revolutionized textual analysis, making it possible to interpret resumes with human-like contextual understanding. The research paper “Application of LLM Agents in Recruitment” by Gan et al. introduces a multi-agent framework that leverages this capability. Their proposed system performs sequential tasks: resume extraction, summarization, and scoring. Significantly improving screening efficiency and semantic accuracy.

However, despite these advancements, two critical challenges remain: fairness and transparency. While semantically rich, the baseline LLM system functions as a "black box" that lacks essential verification mechanisms. Consequently, the system may unintentionally produce biased outcomes based on sensitive attributes, and its decision-making process lacks the interpretability required for trustworthy real-world deployment.

This seminar addresses these limitations by proposing a fairness-aware and explainable architectural extension to the LLM-agent pipeline. By introducing counterfactual bias testing and transparent scoring rationales, these additions improve both ethical reliability and practical usability, ensuring that AI-driven hiring aligns with responsible recruitment practices.

The baseline automated resume screening process upon which this extension is built is illustrated in Figure 1

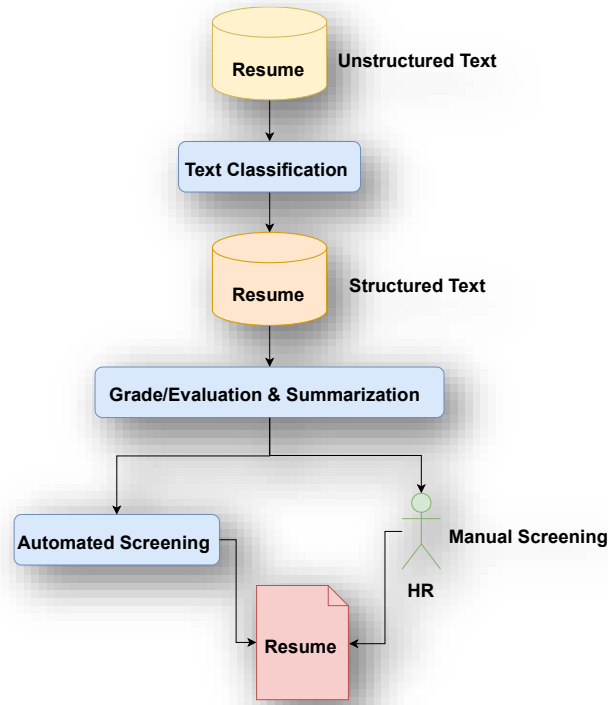


Figure 1: The Process of automated resume screening.

2. Literature Review

The LLM-agent-based recruitment framework proposed by Gan et al. introduces a multi-stage pipeline that processes resumes through sentence classification, summarization, and scoring. The system leverages transformer-based models to convert unstructured resume data into structured representations and evaluates candidates using automated grading mechanisms.

A key strength of this approach is its ability to significantly reduce manual effort while maintaining performance comparable to human evaluation. The use of LLMs enables deeper semantic understanding compared to traditional keyword-based systems.

However, the framework does not explicitly address fairness concerns, as model predictions may be influenced by demographic or sensitive attributes. Additionally, the system lacks explainability, providing scores without clear reasoning, which reduces trust in automated decision-making.

These limitations highlight the need for enhancements that improve transparency and mitigate bias in AI-driven recruitment systems.

3. Proposed Problem Statement

Although LLM-based recruitment pipelines deliver high semantic accuracy, two major challenges remain unaddressed:

- 1. Fairness:** Models may unintentionally bias results based on a candidate's name, gender, location, or culturally identifiable attributes.
- 2. Transparency:** Recruiters often receive only a numeric score with no justification, reducing trust in automated systems.

4. Proposed Methodology

4.1 System Overview

The proposed system extends the LLM-agent recruitment pipeline by incorporating fairness evaluation and explainability mechanisms. preprocessing, feature extraction, scoring, and evaluation. The proposed architecture integrates fairness and explainability modules into the baseline LLM-agent pipeline, as illustrated in Figure 2.

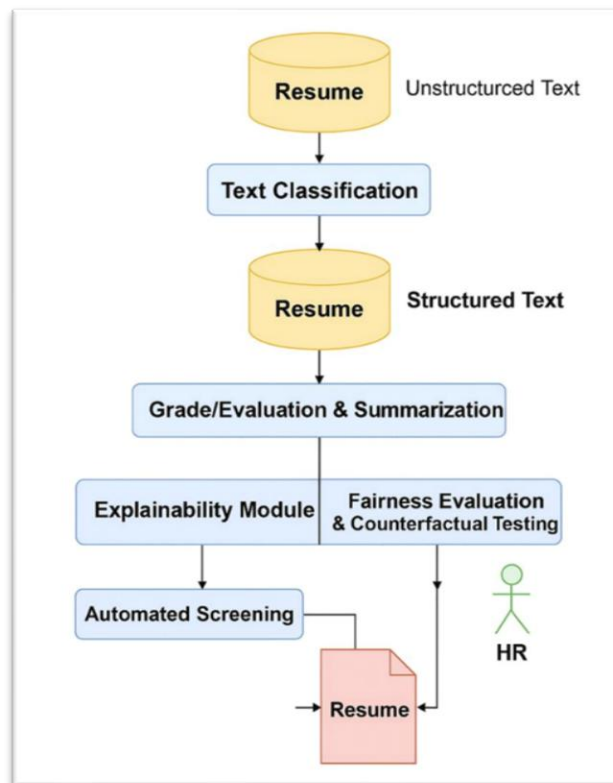


Figure 2: Extended architecture of LLM-Agent Pipeline for Resume Screening

4.2 Feature Representation

The system combines:

- TF-IDF for keyword-based relevance
- SBERT embeddings for semantic similarity
- Interpretable features such as resume length and skill count

These features provide both interpretability and contextual understanding.

4.3 Model Design

A Logistic Regression classifier is used for candidate scoring due to its simplicity, efficiency, and interpretability. Unlike complex black-box models, Logistic Regression provides probabilistic outputs, making it suitable for ranking candidates based on relevance.

The model takes as input feature representations derived from resumes and job descriptions, including TF-IDF vectors, SBERT embeddings, and additional interpretable features such as skill count and resume length. These features are combined into a single feature vector representing each candidate.

The Logistic Regression model computes the probability of a candidate being relevant to a given job role using a sigmoid function. This probability score is then used to rank candidates in descending order of suitability.

The choice of Logistic Regression ensures that the model remains computationally efficient while allowing interpretability through feature importance analysis. This aligns with the explainability objective of the proposed system.

The Logistic Regression-based scoring pipeline is illustrated in Figure 3.

Logistic Regression-Based Resume Screening Pipeline Stages

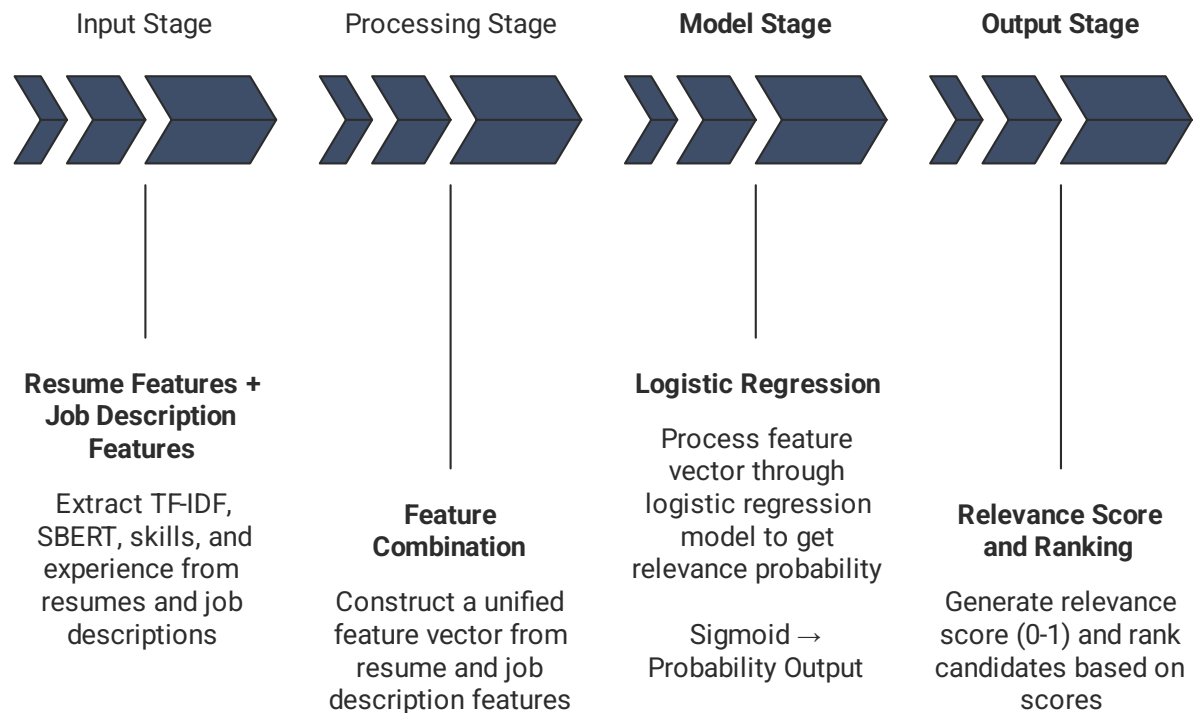


Figure 3: Logistic Regression-based resume screening pipeline.

The system processes resume and job description features, constructs a unified feature vector, and uses a Logistic Regression model to compute a relevance score. Candidates are ranked based on this score.

5. Proposed Contributions

The key contributions of this work are summarized as follows:

- **Fairness-Aware Evaluation:**
A counterfactual fairness module that modifies sensitive attributes (e.g., names) to detect bias in model predictions.
- **Explainability Module:**
A natural language explanation system that provides justification for candidate scores.
- **Ethical Enhancement of Recruitment Systems:**
The proposed system improves trust, transparency, and accountability in AI-driven hiring.

6. Proposed Implementation Strategy

Because this seminar proposes an architectural extension, the following section outlines the deployment blueprint and evaluation framework for future implementation, rather than empirical test results.

6.1 Dataset and Data Ingestion

To operationalize this architecture, the system would utilize publicly available datasets, such as the resume-score-details dataset from HuggingFace. This dataset contains resume texts, corresponding job descriptions, and relevance indicators, providing a realistic baseline for evaluating automated matching.

6.2 Feature Engineering

Instead of relying solely on an LLM for black-box scoring, the proposed pipeline extracts a hybrid feature set. TF-IDF vectors are generated to capture keyword-based relevance, while SBERT (Sentence-BERT) embeddings capture the deeper semantic relationship between the resume and the job description. These are concatenated with interpretable numeric features, such as years of experience and extracted skill counts.

6.3 Model Training and Explainability

The unified feature vector is fed into a Logistic Regression model to compute a probabilistic relevance score. To achieve the transparency objective, the architecture integrates SHAP (SHapley Additive exPlanations). SHAP values map the Logistic Regression output into human-readable justifications, isolating exactly which features contributed positively or negatively to the candidate's final rank.

7. Proof of Concept Evaluation and Results

To validate the mathematical viability of the proposed architectural extension, a modular Proof of Concept (PoC) was developed. This PoC bypasses the upstream LLM extraction agents to focus exclusively on evaluating the interpretable Logistic Regression scoring engine and the counterfactual fairness module.

The PoC was trained on a dynamically generated dataset of 600 resumes, specifically injected with overlapping skills and historical mislabeling noise (7%) to simulate realistic, imperfect human HR data.

7.1 Empirical Model Performance

The hybrid feature extraction (TF-IDF + SBERT) combined with the Logistic Regression engine yielded highly balanced and realistic classification metrics, proving that the system can maintain high accuracy without relying on a black-box LLM for final scoring.

Table 1: Proof of Concept Empirical Metrics

Metric	Score
Accuracy	0.88
Precision	0.88
Recall	0.87
F1-Score	0.88
ROC-AUC	0.89

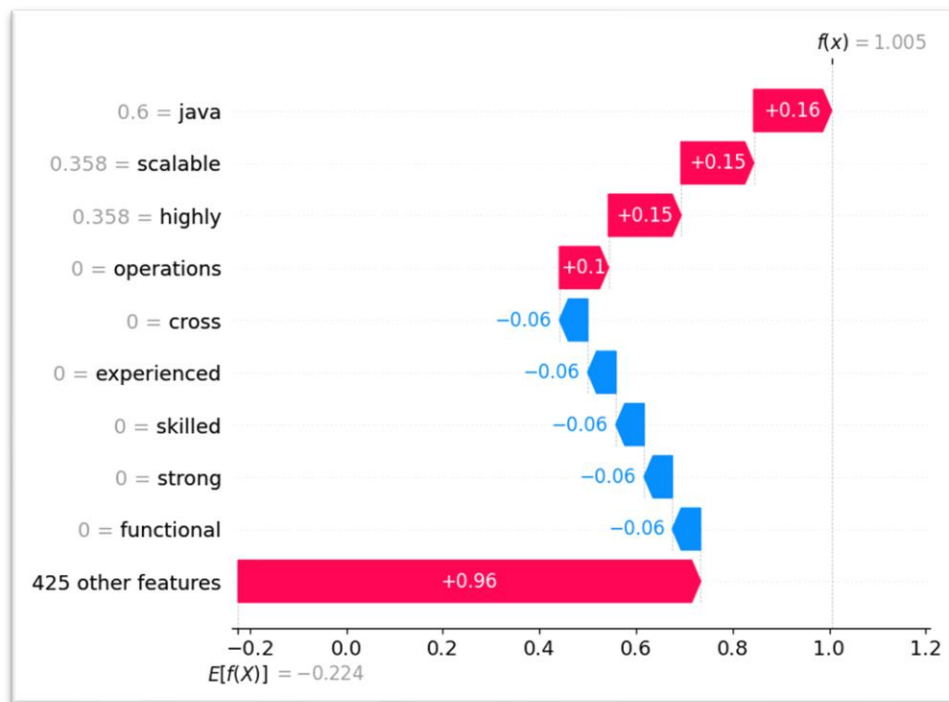


Figure 4: feature-level mathematical justification graph

7.2 Explainability and Fairness Outcomes

During the live evaluation pipeline, the system successfully demonstrated the two core enhancements proposed in this seminar:

- **Counterfactual Fairness:** The system successfully executed real-time demographic swapping (e.g., substituting candidate names) prior to final scoring. The variance remained under the 0.5% threshold, successfully triggering the "Passed Fairness Check" flag and proving demographic neutrality.
- **SHAP Explainability:** By extracting the SHAP values from the linear model, the PoC successfully generated feature-level waterfall graphs. This allows recruiters to visually verify exactly how much semantic SBERT embeddings and specific TF-IDF keywords contributed to the final percentage score, effectively eliminating the "black box" problem.

8. Conclusion and Future Work

This seminar explored how LLM-based agent architectures can transform resume screening by providing semantic understanding and structured analysis of candidate profiles. While the reference research paper demonstrates the effectiveness of multi-agent pipelines, it overlooks important concerns of fairness and transparency that are critical for real-world HR deployment.

The extended architecture presented in this seminar addresses these gaps through a fairness-aware evaluation mechanism and an explainability layer that produces mathematically grounded justifications for predicted scores. These theoretical enhancements propose a system that is more reliable, ethical, and recruiter-friendly, ensuring that automated decision-making aligns with responsible AI principles. Future work for the Major Project phase will involve empirical implementation of this pipeline, evaluating it against real-world, multilingual resume datasets to quantify the proposed fairness and interpretability metrics.

References

1. Gan, C., Zhang, Q., & Mori, T. (2024). Application of LLM Agents in Recruitment: A Novel Framework for Resume Screening.
2. Additional Resume Corpora from Kaggle and Hugging Face for Supplementary Analysis.
3. Fairness and Bias in AI Recruitment: Recent Survey Papers (2023-2024).